

ELECTRONIC SUPPLEMENTARY MATERIAL

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Uncertainty of UV-Vis spectrophotometric and FAAS analysis for determination of iron in pharmaceutical product

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Composition of the Bio-Multi[®] formulation

Bio-Multi[®] (one tablet contained 10 mg Fe), produced by Pharma Nord ApS, from local pharmacy were used for Fe determination as sample. Bio-Multi[®] is a multivitamin with minerals. One coated tablet contains 10 vitamins: 60 mg ascorbic acid (vitamin C), 3.6 mg β -carotene (vitamin A), 4 mg d- α -tocopheryl acetate (vitamin E), 9 mg nicotine amide (niacin), 7.5 μ g cholecalciferol (vitamin D), 1.5 μ g cyanocobalamine (vitamin B₁₂), 1 mg riboflavin (vitamin B₂), 0.9 mg pyridoxine hydrochloride (vitamin B₆), 0.6 mg thiamine hydrochloride (vitamin B₁), 300 μ g folic acid and 6 minerals: 100 mg calcium (mild lime and dicalcium phosphate), 100 mg magnesium (magnesia), 10 mg iron (ferrous gluconate), 5 mg zinc (zinc gluconate), 100 μ g iodine (potassium iodide), 60 μ g selenium (leaven, which is enriched with sodiumselenite). Anticakings are microcrystalline cellulose, silica, talcum and magnesium stearate. Stabilizers are hypromellose and polyvinylpyrrolidone. Food colours are titania and Marigold extract.

Table S1. Instrumental conditions for measurement of iron by flame atomic absorption spectrometry

Parameter	Value (Element Fe)
Wavelength (nm)	248.3
Slit (nm)	0.2
Light source	Hollow cathode lamp (Co-Cr-Cu-Fe-Mn-Ni)
Lamp current (mA)	10
Flame, flow setting (l/min)	Air (3.0), acetylene (1.1)
Integration time (s)	3.0

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Uncertainty of weighing and volumetric operations

1. Weighing

u(m, rep)=	0.0001	g	u(weighing)=	1E-04	g
u(m, drift)=	0.0001	g			
u(m, roun)=	2.887E-05				

2. Volumetric flasks

50 ml flask (± 0.09 ml)	100 ml flask (± 0.15 ml)	250 ml flask (± 0.15 ml)
u(cal)= 0.0520 ml	u(cal)= 0.0866 ml	u(cal)= 0.0866 ml
u(fill)= 0.0520 ml	u(fill)= 0.0520 ml	u(fill)= 0.0520 ml
u(temp)= 0.0121 ml	u(temp)= 0.0242 ml	u(temp)= 0.0606 ml
u(V50)= 0.0745 ml	u(V100)= 0.1039 ml	u(V250)= 0.1178 ml

3. Pipettes

1-ml pipet (± 0.01 ml)	2-ml pipet (± 0.015 ml)	5-ml pipet (± 0.010 ml)
u(cal)= 0.0058 ml	u(cal)= 0.0087 ml	u(cal)= 0.0058 ml
u(rep)= 0.0040 ml	u(rep)= 0.0080 ml	u(rep)= 0.0200 ml
u(temp)= 0.0002 ml	u(temp)= 0.0005 ml	u(temp)= 0.0012 ml
u(V1)= 0.0070 ml	u(V2)= 0.0118 ml	u(V5)= 0.0209 ml
		df= 9

10-ml pipet (± 0.03 ml)	25-ml pipet (± 0.03 ml)
u(cal)= 0.0173 ml	u(cal)= 0.0173 ml
u(rep)= 0.0400 ml	u(rep)= 0.1000 ml
u(temp)= 0.0024 ml	u(temp)= 0.0061 ml
u(V10)= 0.0437 ml	u(V25)= 0.1017 ml
df= 9	df= 9

4. Solutions

Stock solution 1: Solution of iron 1000 mg/l

Molar mass of Mohr's salt $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$

(Source: NIST Physical Reference Data)

Element	repeater	M (g/mol)
N	2	14.0067
H	20	1.00794
Fe	1	55.845
S	2	32.065
O	14	15.9994

M (Mohr's salt) = **392.139**

Purity of Mohr's salt (P_M) = 99,5% = 0.995

$u(P_M) = 0.0029$

Mass of Mohr's salt = **0.7002** g

$C(\text{Fe}) = (m_M \cdot M_{\text{Fe}} \cdot P_M \cdot 1000) / (M_M \cdot V_{100})$

$C(\text{Fe}) = \mathbf{992.18}$ mg/l

	Unit	Value	Standard deviation	
			absolute	relative (10^{-4})
m_M	g	0.7002	0.00014	2.06
P_M		0.995	0.0029	29.01
V_{100}	l	0.1	0.00010	10.4

$u(C(\text{Fe}))/C(\text{Fe}) = 0.0031$

$u(C(\text{Fe})) = \mathbf{3.064}$ mg/l

Calibration solution: Fe 0.2 µg/ml

	Unit	Value	Standard deviation	
			absolute	relative (10 ⁴)
C (Fe)	mg/l	992.18	3.06430	30.9
V10	ml	10	0.000	0.0
V10	ml	10	0.000	0.0
V1	ml	1	0.007	70.3
V100	ml	100	0.104	10.4
V50	ml	50	0.074	14.9

The uncertainty of the two 10 ml pipetting operations (stock solutions 1 and 2) is taken into account by f1 and f2 when calculating u(C(sample))

$$C(\sim 0,2\mu\text{g/ml}) = 10\text{ml} \cdot C_M \cdot 10\text{ml} \cdot 1\text{ml} / (100\text{ml} \cdot 100\text{ml} \cdot 50\text{ml})$$

$$C(\sim 0,2\mu\text{g/ml}) = \mathbf{0.198}$$

$$u(C(\sim 0,2\mu\text{g/ml})) / C(\sim 0,2\mu\text{g/ml}) = 0.0080$$

$$u(\sim 0,2\mu\text{g/ml}) = \mathbf{0.00158} \quad \mu\text{g/ml}$$

Explanation:

Stock solution 2: 10 ml stock solution 1 were transferred into 100 ml flask

Stock solution 3: 10 ml stock solution 2 were transferred into 100 ml flask

Standard solution (0.2 µg/ml): 1 ml stock solution 3 were transferred into 50 ml flask

Calibration solution: Fe 0.4 µg/ml

	Unit	Value	Standard deviation	
			absolute	relative (10 ⁴)
C (Fe)	mg/l	992.18	3.06430	30.9
V10	ml	10	0.000	0.0
V10	ml	10	0.000	0.0
V2	ml	2	0.012	59.0
V100	ml	100	0.104	10.4
V50	ml	50	0.074	14.9

The uncertainty of the two 10 ml pipetting operations (stock solutions 1 and 2) is taken into account by f1 and f2 when calculating u(C(sample))

$$C(\sim 0,4\mu\text{g/ml}) = 10\text{ml} \cdot C_M \cdot 10\text{ml} \cdot 2\text{ml} / (100\text{ml} \cdot 100\text{ml} \cdot 50\text{ml})$$

$$C(\sim 0,4\mu\text{g/ml}) = \mathbf{0.397}$$

$$u(C(\sim 0,4\mu\text{g/ml})) / C(\sim 0,4\mu\text{g/ml}) = 0.0070$$

$$u(\sim 0,4\mu\text{g/ml}) = \mathbf{0.00277} \quad \mu\text{g/ml}$$

Calibration solution: Fe 1.0 µg/ml

	Unit	Value	Standard deviation	
			absolute	relative (10 ⁴)
C (Fe)	mg/l	992.18	3.06430	30.9
V10	ml	10	0.000	0.0
V10	ml	10	0.000	0.0
V5	ml	5	0.021	41.7
V100	ml	100	0.104	10.4
V50	ml	50	0.074	14.9

The uncertainty of the two 10 ml pipetting operations (stock solutions 1 and 2) is taken into account by f1 and f2 when calculating u(C(sample))

$$C (\sim 1.0 \mu\text{g/ml}) = 10 \text{ ml} * C_M * 10 \text{ ml} * 5 \text{ ml} / (100 \text{ ml} * 100 \text{ ml} * 50 \text{ ml})$$

$$C (\sim 1.0 \mu\text{g/ml}) = \mathbf{0.992}$$

$$u(C (\sim 1.0 \mu\text{g/ml})) / C (\sim 1.0 \mu\text{g/ml}) = 0.0056$$

$$u (\sim 1.0 \mu\text{g/ml}) = \mathbf{0.00555} \quad \mu\text{g/ml}$$

Calibration solution: Fe 2.0 µg/ml

	Unit	Value	Standard deviation	
			absolute	relative (10 ⁴)
C (Fe)	mg/l	992.18	3.06430	30.9
V10	ml	10	0.000	0.0
V10	ml	10	0.000	0.0
V10	ml	10	0.044	43.7
V100	ml	100	0.104	10.4
V50	ml	50	0.074	14.9

The uncertainty of the two 10 ml pipetting operations (stock solutions 1 and 2) is taken into account by f1 and f2 when calculating u(C(sample))

$$C (\sim 2.0 \mu\text{g/ml}) = 10 \text{ ml} * C_M * 10 \text{ ml} * 10 \text{ ml} / (100 \text{ ml} * 100 \text{ ml} * 50 \text{ ml})$$

$$C (\sim 2.0 \mu\text{g/ml}) = \mathbf{1.984}$$

$$u(C (\sim 2.0 \mu\text{g/ml})) / C (\sim 2.0 \mu\text{g/ml}) = 0.0057$$

$$u (\sim 2.0 \mu\text{g/ml}) = \mathbf{0.01139} \quad \mu\text{g/ml}$$

Calibration solution: Fe 3.0 µg/ml

	Unit	Value	Standard deviation	
			absolute	relative (10 ⁴)
C (Fe)	mg/l	992.18	3.06430	30.9
V10	ml	10	0.000	0.0
V10	ml	10	0.000	0.0
V5	ml	5	0.02085	41.7
V10	ml	10	0.044	43.7
V100	ml	100	0.104	10.4
V50	ml	50	0.074	14.9

The uncertainty of the two 10 ml pipetting operations (stock solutions 1 and 2) is taken into account by f1 and f2 when calculating u(C(sample))

$$C (\sim 3.0 \mu\text{g/ml}) = 10 \text{ ml} * C_M * 10 \text{ ml} * (10 \text{ ml} + 5 \text{ ml}) / (100 \text{ ml} * 100 \text{ ml} * 50 \text{ ml})$$

$$C (\sim 3.0 \mu\text{g/ml}) = \mathbf{2.977}$$

$$u(C (\sim 3.0 \mu\text{g/ml})) / C (\sim 3.0 \mu\text{g/ml}) = 0.0071$$

$$u (\sim 3.0 \mu\text{g/ml}) = \mathbf{0.02112} \quad \mu\text{g/ml}$$

Calibration solution: Fe 4.0 µg/ml

	Unit	Value	Standard deviation	
			absolute	relative (10 ⁴)
C (Fe)	mg/l	992.18	3.06430	30.9
V10	ml	10	0.000	0.0
V10	ml	10	0.000	0.0
V10	ml	10	0.044	43.7
V100	ml	100	0.104	10.4
V50	ml	50	0.074	14.9

The uncertainty of the two 10 ml pipetting operations (stock solutions 1 and 2) is taken into account by f1 and f2 when calculating u(C(sample))

$$C (\sim 4.0 \mu\text{g/ml}) = 10 \text{ ml} * C_M * 10 \text{ ml} * (10 \text{ ml} + 10 \text{ ml}) / (100 \text{ ml} * 100 \text{ ml} * 50 \text{ ml})$$

$$C (\sim 4.0 \mu\text{g/ml}) = \mathbf{3.969}$$

$$u(C (\sim 4.0 \mu\text{g/ml})) / C (\sim 4.0 \mu\text{g/ml}) = 0.0072$$

$$u (\sim 4.0 \mu\text{g/ml}) = \mathbf{0.02863} \quad \mu\text{g/ml}$$

Uncertainty of UV-Vis Spectrophotometry

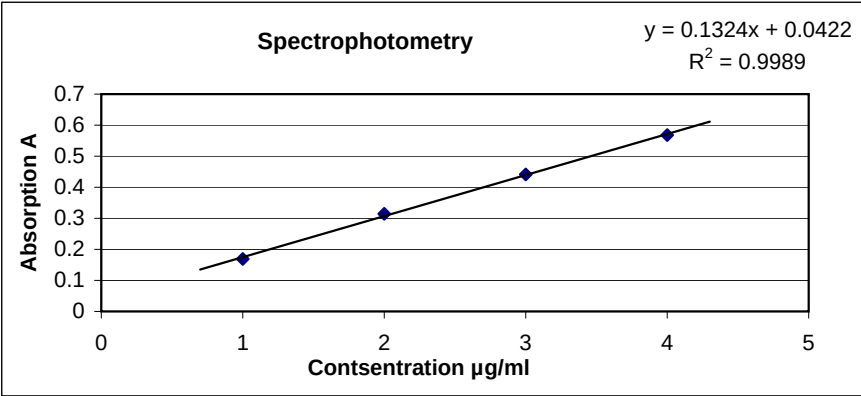
Uncerativity of instrument signal for solutions of calibration curve (AU)

C (µg/ml)	A1	A2	A3	A4	A5	Average A	s	arith.aver.s	squared
Blank sol.	0.001	0.002	0.002	0.002	0.002	0.0018			
1	0.171	0.171	0.171	0.171	0.171	0.1692	0	0	0
2	0.316	0.316	0.316	0.316	0.316	0.3142	0	0	0
3	0.443	0.443	0.443	0.444	0.444	0.4416	0.000548	0.00024495	6E-08
4	0.570	0.570	0.570	0.570	0.570	0.5682	0	0	0

6E-08 sum

Repeatability	0.000122
Rounding:	0.000289

Combined u: 0.000314 AU



Uncertainty of absorbance of the sample solution

Sample	A1	A2	A3	A4	A5	Average A	s	u(Avg A,rep)	Drift	u(Avg A,Drift)
Bio-Multi	0.344	0.344	0.343	0.344	0.344	0.342	0.000447	0.00020	0.0015	0.000866025

max nonlin	u(Avg A, nonlin)	u(Avg A, Phen)	u(Avg A, Hydroq)	u(Avg A, Na-citr)	u(Avg A, round)
0.007	0.00404	0.00058	0.00058	0.00058	0.000289

Uncertainty of sample solution absorbance:

$$u(A(\text{sample})) = u(\text{Avg A}) = \mathbf{0.00427 \quad AU}$$

$$\text{No of eff df of } A(\text{Sample}) = 5$$

The factors f1 and f2 take into account the uncertainties of the two common pipetting operations when preparing the standard solutions. These two pipetting operations affect C1..C4 te same way and thus introduce correlation between these concentrations. The used approach allows to automatically take this correlation into account.

The effective number of degrees of freedom of A(sample) is fully determined by the nonlinearity contribution. Formally rectangularly distributed quantities have an infinite number of degrees of freedom. However, this is unjustified in our case A more sound approach is to estimate the eff df number as the number of such calibration lines constructed during the experiments minus 1.

Kragten approach is used for calculating the uncertainty of C(sample):

	value	u	C1	C2	C3	C4	f1	f2	A1	A2	A3	A4	A(sample)
C1	0.9922	0.0056	0.9977	0.9922	0.9922	0.9922	0.9922	0.9922	0.9922	0.9922	0.9922	0.9922	0.9922
C2	1.9844	0.0114	1.9844	1.9958	1.9844	1.9844	1.9844	1.9844	1.9844	1.9844	1.9844	1.9844	1.9844
C3	2.9765	0.0211	2.9765	2.9765	2.9977	2.9765	2.9765	2.9765	2.9765	2.9765	2.9765	2.9765	2.9765
C4	3.9687	0.0286	3.9687	3.9687	3.9687	3.9973	3.9687	3.9687	3.9687	3.9687	3.9687	3.9687	3.9687
f1	1	0.004366	1.0000	1.0000	1.0000	1.0000	1.0044	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
f2	1	0.004366	1.0000	1.0000	1.0000	1.0000	1.0000	1.0044	1.0000	1.0000	1.0000	1.0000	1.0000
A1	0.1692	0.000314	0.1692	0.1692	0.1692	0.1692	0.1692	0.1692	0.1695	0.1692	0.1692	0.1692	0.1692
A2	0.3142	0.000314	0.3142	0.3142	0.3142	0.3142	0.3142	0.3142	0.3142	0.3145	0.3142	0.3142	0.3142
A3	0.4416	0.000314	0.4416	0.4416	0.4416	0.4416	0.4416	0.4416	0.4416	0.4416	0.4419	0.4416	0.4416
A4	0.5682	0.000314	0.5682	0.5682	0.5682	0.5682	0.5682	0.5682	0.5682	0.5682	0.5682	0.5685	0.5682
A(sample)	0.3420	0.00427	0.3420	0.3420	0.3420	0.3420	0.3420	0.3420	0.3420	0.3420	0.3420	0.3420	0.3463

$$C(\text{sample}) = \mathbf{2.2460}$$

$$u(C(\text{sample})) = \mathbf{0.0357}$$

	2.2477	2.2491	2.2507	2.2510	2.255766	2.2557656	2.2452	2.2453	2.2454	2.2455	2.2779
Delta:	0.0018	0.0031	0.0048	0.0051	0.0098	0.0098	-0.0008	-0.0006	-0.0005	-0.0004	0.0320
Delta ² :	3.134E-06	9.876E-06	2.287E-05	2.578E-05	9.614E-05	9.614E-05	5.691E-07	4.134E-07	2.827E-07	1.768E-07	1.022E-03
Summa:	0.0012772										
Index:	0.25%	0.77%	1.79%	2.02%	7.53%	7.53%	0.04%	0.03%	0.02%	0.01%	80.00%

Average tablet mass: $m_{\text{tablet_average}}:$

m1 = 0.8866 g
m2 = 0.8904 g
m3 = 0.9010 g
m4 = 0.8893 g
m5 = 0.9049 g
m6 = 0.9063 g
m7 = 0.9193 g
m8 = 0.8953 g
m9 = 0.9149 g
m10 = 0.8753 g
m11 = 0.9072 g
m12 = 0.9070 g
m13 = 0.8966 g
m14 = 0.8962 g
m15 = 0.8936 g
m16 = 0.8947 g
m17 = 0.9043 g
m18 = 0.8857 g
m19 = 0.9066 g
m20 = 0.8874 g

 $m_{\text{tablet_average}} = 0.89813 \text{ g}$ $m_{\text{sample}} = 0.8853 \text{ g}$

Content of iron in tablet

$$m_{\text{Fe_per_tablet}} = \frac{C(\text{sample}) \cdot V_{50} \cdot V_{100} \cdot V_{250} \cdot m_{\text{tablet_average}}}{V_{\text{sample}} \cdot V_{25} \cdot m_{\text{sample}} \cdot 1000}$$

m_Fe_per_tablet = 11.393

	Unit	Value	Standard deviation		v_i	Contributions
			absolute	relative (10 ⁴)		
Csample	µg/ml	2.2460	0.0357	159.1	5	57.51%
m _{tablet_average}	g	0.89813	0.00014	1.6		0.58%
m _{sample}	g	0.8853	0.00014	1.6		0.59%
V50	ml	50	0.0745	14.9		5.38%
V100	ml	100	0.1039	10.4		3.75%
V250	ml	250	0.1178	4.7		1.70%
V10	ml	10	0.0437	43.7	9	15.78%
V25	ml	25	0.1017	40.7	9	14.70%

u(m_Fe_per_tablet) / m_Fe_per_tablet = 0.0171
 u(m_Fe_per_tablet) = **0.195**

m_Fe_per_tablet	u _c	U(95.45%)	U _{rel.} %
11.393	0.195	0.490	4.30

The Welch-Satterthwaite

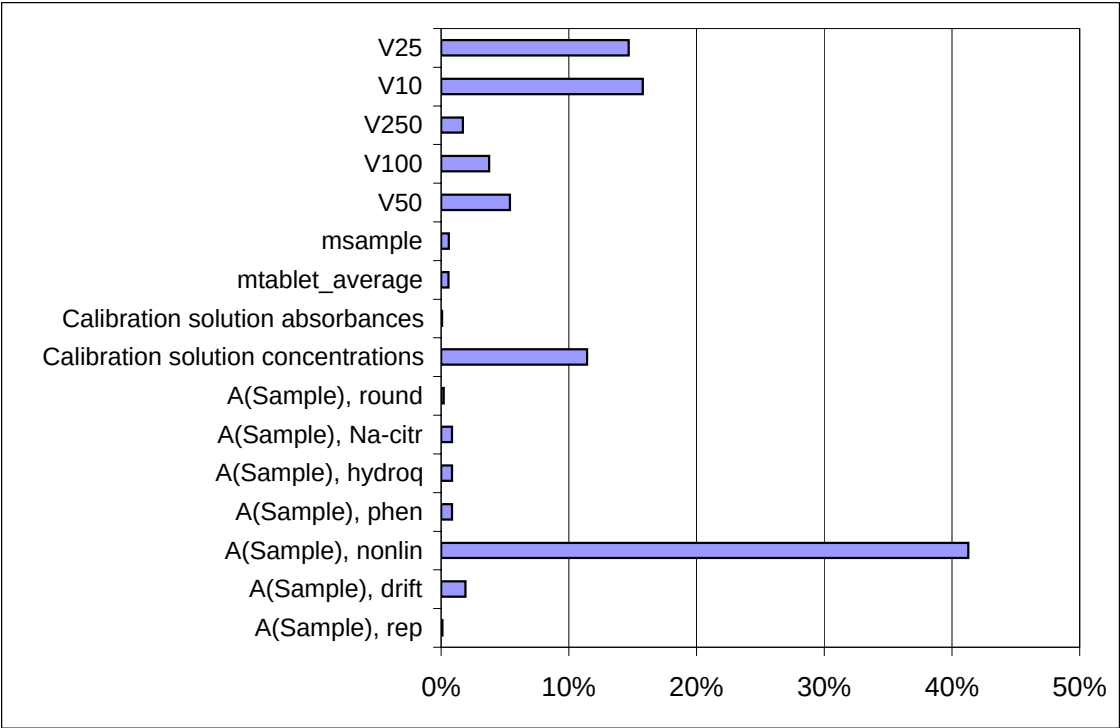
formula:

$$v_{\text{eff}} = \frac{u_c^4}{\sum_i \frac{u_i^4}{v_i}}$$

$v_{\text{eff}} = 6.5$
 $k(95.45\%) = 2.52$

Result:				
11.39	±	0.49	mg per tablet	k(95.45%) = 2.52

Uncertainty budget:



Contributions of C(sample) components:

A(Sample):	46.01%
Calibration solution concentrations:	11.43%
Calibration solution absorbances:	0.06%

Components of A(sample):

	Index	u	u squared
Rep	0.10%	0.00020	4E-08
Drift	1.90%	0.000866025	7.5E-07
Nonlin	41.28%	0.00404	1.63E-05
Phen.	0.84%	0.00058	3.33E-07
Hydroq.	0.84%	0.00058	3.33E-07
Na-citr.	0.84%	0.00058	3.33E-07
Rounding	0.21%	0.00029	8.33E-08
	sumsq:		1.82E-05

Uncertainty components	contributions (mg)	contributions (%)
A (Sample), rep	0.02	0.1%
A (Sample), drift	0.07	1.9%
A (Sample), nonlin	0.31	41.3%
A (Sample), phen	0.04	0.8%
A (Sample), hydroq	0.04	0.8%
A (Sample), Na-citr	0.04	0.8%
A (Sample), round	0.02	0.2%
Calibration solution concentrations	0.17	11.4%
Calibration solution absorbances	0.01	0.1%
m tablet_average	0.04	0.6%
msample	0.04	0.6%
V50	0.11	5.4%
V100	0.09	3.8%
V250	0.06	1.7%
V10	0.19	15.8%
V25	0.19	14.7%

UV-Vis residual analysis

SUMMARY OUTPUT

<i>Regression Statistics</i>	
Multiple R	0.99943748
R Square	0.99887527
Adjusted R Square	0.9983129
Standard Error	0.00702681
Observations	4

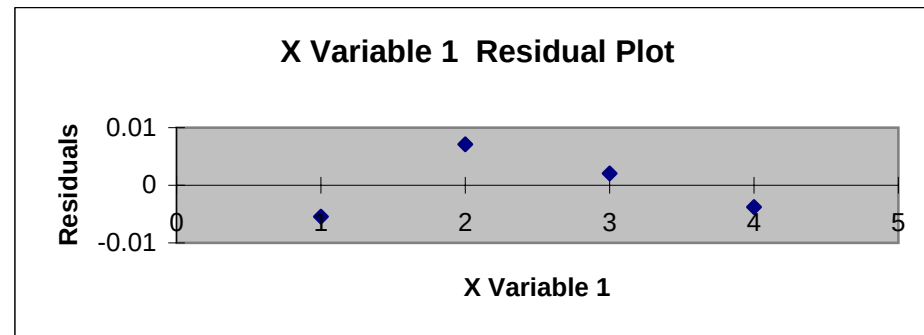
ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	0.087701768	0.087701768	1776.202	0.000562524
Residual	2	0.000098752	0.000049376		
Total	3	0.08780052			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	0.0422	0.008606044	4.903530368	0.039163	0.005171154	0.07922885	0.005171154	0.079228846
X Variable 1	0.13244	0.003142483	42.1450159	0.000563	0.118918977	0.14596102	0.118918977	0.145961023

RESIDUAL OUTPUT

<i>Observation</i>	<i>Predicted Y</i>	<i>Residuals</i>	<i>Standard Residuals</i>
1	0.17464	-0.00544	-0.948170801
2	0.30708	0.00712	1.240988254
3	0.43952	0.00208	0.362535894
4	0.57196	-0.00376	-0.655353348

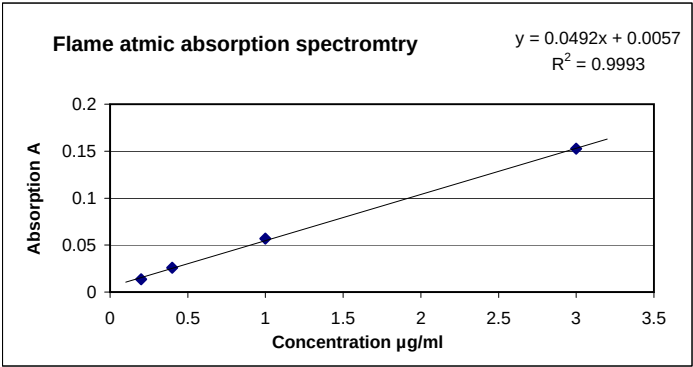


Uncertainty of Flame atomic absorption spectrometry

Uncerativity of instrument signal for solutions of calibration curve (AU)

C (µg/ml)	A1	A2	A3	A4	A5	Average A	s	arith.aver.s	squared
0.2	0.013	0.013	0.014	0.014	0.014	0.0136	0.00054772	0.000244949	6E-08
0.4	0.025	0.026	0.026	0.026	0.027	0.0260	0.00070711	0.000316228	1E-07
1	0.057	0.056	0.057	0.057	0.057	0.0568	0.00044721	0.00020000	4E-08
3	0.153	0.152	0.153	0.153	0.152	0.1526	0.00054772	0.000244949	6E-08
									2.6E-07 sum
Repeatability:									0.000255
Rounding:									0.0002887

Combined u: 0.0003851 AU



Uncertainty of absorbance of the sample solution

Sample	A1	A2	A3	A4	A5	Average A	s	u(Avg A,rep)	Drift	u(Avg A,Drift)
Bio-Multi	0.064	0.065	0.064	0.065	0.065	0.0646	0.00054772	0.00024	0.0018	0.00103923

max nonlin	u(Avg A, nonlin)	u(Avg A, HCl)	u(Avg A, Round)
0.002	0.00115	0.00058	0.00029

Uncertainty of sample solution absorbance:

$$u(A(\text{sample})) \equiv u(\text{Avg A}) = \mathbf{0.00170 \quad AU}$$

$$\text{No of eff df of A(Sample)} = 9.89$$

The factors f1 and f2 take into account the uncertainties of the two common pipetting operations when preparing the standard solutions. These two pipetting operations affect C1..C4 te same way and thus introduce correlation between these concentrations. The used approach allows to automatically take this correlation into account.

The effective number of degrees of freedom of A(sample) is fully determined by the nonlinearity contribution. Formally rectangularly distributed quantities have an infinite number of degrees of freedom. However, this is unjustified in our case A more sound approach is to estimate the eff df number as the number of such calibration lines constructed during the experiments minus 1.

Kragten approach is used for calculating the uncertainty of C(sample):

	value	u	C1	C2	C3	C4	f1	f2	A1	A2	A3	A4	A(sample)
C1	0.1984	0.0016	0.2000	0.1984	0.1984	0.1984	0.1984	0.1984	0.1984	0.1984	0.1984	0.1984	0.1984
C2	0.3969	0.0028	0.3969	0.3996	0.3969	0.3969	0.3969	0.3969	0.3969	0.3969	0.3969	0.3969	0.3969
C3	0.9922	0.0056	0.9922	0.9922	0.9977	0.9922	0.9922	0.9922	0.9922	0.9922	0.9922	0.9922	0.9922
C4	2.9765	0.0211	2.9765	2.9765	2.9765	2.9977	2.9765	2.9765	2.9765	2.9765	2.9765	2.9765	2.9765
f1	1	0.004366	1	1	1	1	1.00437	1	1	1	1	1	1
f2	1	0.004366	1	1	1	1	1	1.00437	1	1	1	1	1
A1	0.0136	0.000385	0.0136	0.0136	0.0136	0.0136	0.0136	0.0136	0.0140	0.0136	0.0136	0.0136	0.0136
A2	0.026	0.000385	0.0260	0.0260	0.0260	0.0260	0.0260	0.0260	0.0260	0.0264	0.0260	0.0260	0.0260
A3	0.0568	0.000385	0.0568	0.0568	0.0568	0.0568	0.0568	0.0568	0.0568	0.0568	0.0572	0.0568	0.0568
A4	0.1526	0.000385	0.1526	0.1526	0.1526	0.1526	0.1526	0.1526	0.1526	0.1526	0.1526	0.1530	0.1526
A(sample)	0.0646	0.00170	0.0646	0.0646	0.0646	0.0646	0.0646	0.0646	0.0646	0.0646	0.0646	0.0646	0.0663

$$C(\text{sample}) = \mathbf{1.1884}$$

$$u(C(\text{sample})) = \mathbf{0.0358}$$

Delta:	0.0004	0.0007	0.0014	0.0057	0.0052	0.0052	-0.0019	-0.0019	-0.0019	-0.0021	0.0343
Delta ² :	1.450E-07	4.516E-07	1.899E-06	3.208E-05	2.692E-05	2.692E-05	3.513E-06	3.568E-06	3.734E-06	4.315E-06	1.177E-03
Summa:	0.0012808										
Index:	0.01%	0.04%	0.15%	2.50%	2.10%	2.10%	0.27%	0.28%	0.29%	0.34%	91.92%

Average tablet mass: $m_{\text{tablet_average}} =$

m1 = 0.8866 g

m2 = 0.8904 g

m3 = 0.9010 g

m4 = 0.8893 g

m5 = 0.9049 g

m6 = 0.9063 g

m7 = 0.9193 g

m8 = 0.8953 g

m9 = 0.9149 g

m10 = 0.8753 g

m11 = 0.9072 g

m12 = 0.9070 g

m13 = 0.8966 g

m14 = 0.8962 g

m15 = 0.8936 g

m16 = 0.8947 g

m17 = 0.9043 g

m18 = 0.8857 g

m19 = 0.9066 g

m20 = 0.8874 g

 $n_{\text{tablet_average}} = 0.89813 \text{ g}$ $m_{\text{sample}} = 0.8991 \text{ g}$

Content of iron in tablet

$$m_{\text{Fe_per_tablet}} = \frac{C(\text{sample}) \cdot V_{50} \cdot V_{100} \cdot V_{250} \cdot m_{\text{tablet_average}}}{V_{\text{sample}} \cdot V_{25} \cdot m_{\text{sample}} \cdot 1000}$$

m_Fe_per_tablet = 11.872

	Unit	Value	Standard deviation		v _i	Contributions
			absolute	relative (10 ⁴)		
Csample	µg/ml	1.1884	0.0358	301.1	9.89	72.26%
m _{tablet_average}	g	0.8981	0.00014	1.6		0.39%
m _{sample}	g	0.8991	0.00014	1.6		0.39%
V50	ml	50	0.0745	14.9		3.57%
V100	ml	100	0.1039	10.4		2.49%
V250	ml	250	0.1178	4.7		1.13%
V5	ml	5	0.0209	41.7	9	10.01%
V25	ml	25	0.1017	40.7	9	9.76%

u(m_Fe_per_tablet) / m_Fe_per_tablet =

0.0307

The Welch-Satterthwaite

u(m_Fe_per_tablet) = **0.365**

formula:

m_Fe_per_tablet	u _C	U(95.45%)	U _{rel} (k=2), %
11.872	0.365	0.833	7.02

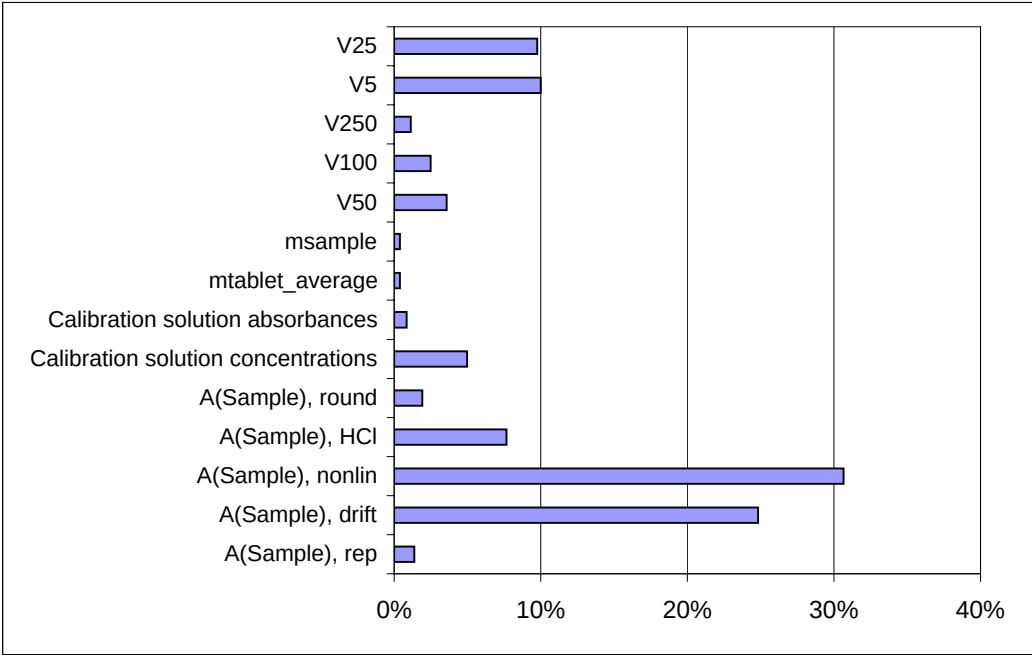
$$v_{\text{eff}} = \frac{u_c^4}{\sum_i \frac{u_i^4}{v_i}}$$

v_{eff} = 10.6

k(95.45%) = 2.28

Result:					
11.87	±	0.83	mg per tablet	k (95.45%)=	2.28

Uncertainty budget:



Contributions of C(sample) components:	
A(sample):	66.42%
Calibration solution concentrations:	4.99%
Calibration solution absorbances:	0.85%

Components of A(sample):			
	Index	u	u squared
Rep	1.38%	0.00024	6E-08
Drift	24.82%	0.00103923	0.00000108
Nonlin	30.64%	0.00115	1.33333E-06
HCl	7.66%	0.00058	3.33333E-07
Rounding	1.92%	0.00029	8.33333E-08
	sumsq:		2.89E-06

Uncertainty components	contributions (mg)	contributions (%)
A (Sample), rep	0.10	1.4%
A (Sample), drift	0.42	24.8%
A (Sample), nonlin	0.46	30.6%
A (Sample), HCl	0.23	7.7%
A (Sample), round	0.12	1.9%
Calibration solution concentrations	0.19	5.0%
Calibration solution absorbances	0.08	0.9%
m tablet_average	0.05	0.4%
m sample	0.05	0.4%
V50	0.16	3.6%
V100	0.13	2.5%
V250	0.09	1.1%
V5	0.26	10.0%
V25	0.26	9.8%

FAAS residual analysis

SUMMARY OUTPUT

<i>Regression Statistics</i>	
Multiple R	0.99965312
R Square	0.99930636
Adjusted R Squ	0.99895954
Standard Error	0.0020293
Observations	4

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	0.011865474	0.011865474	2881.337	0.000346881
Residual	2	8.23609E-06	4.11804E-06		
Total	3	0.01187371			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	0.00571731	0.001462429	3.909461901	0.059636	-0.00057502	0.012009641	-0.000575018	0.012009641
X Variable 1	0.04915886	0.000915809	53.67808461	0.000347	0.04521845	0.053099269	0.04521845	0.053099269

RESIDUAL OUTPUT

<i>Observation</i>	<i>Predicted Y</i>	<i>Residuals</i>	<i>Standard Residuals</i>
1	0.01554908	-0.001949	-1.176333706
2	0.02538086	0.000619	0.373673403
3	0.05487617	0.001924	1.161091765
4	0.15319389	-0.000594	-0.358431462

